Multiple Recursion: Generate Strings Using A, B and C

Aim:
To generate all possible strings of length n using the characters A, B, and C using **multiple recursion**.

Algorithm

1. Read the required string length n.
2. At each position, choose A.
3. Recursively generate the remaining positions.
4. Choose B and recursively continue.
5. Choose C and recursively continue.
6. When the required length is reached, display the string.

Expected Input
Enter length: 2

Expected Output
AA
AB
AC
BA
BB
BC
CA
CB
CC

Number of strings
For length 3:

Divide and Conquer: Quick Sort

Aim:
To sort an array using **recursive Quick Sort based on divide-and-conquer**.

Algorithm

1. Read the array.
2. Select a pivot.
3. Partition the array into elements smaller and larger than the pivot.
4. Recursively sort the left subarray.
5. Recursively sort the right subarray.
6. Combine the partitions through the pivot.
7. Display the sorted array.

Expected Input
Enter size: 7

Elements:
38 12 45 7 19 31 25

Expected Output
Sorted array:
7 12 19 25 31 38 45